

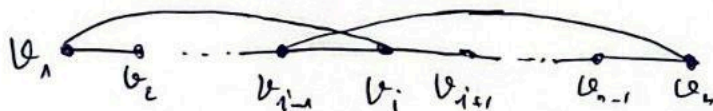
(3)

Hamiltonian Graphs and Diagraphs:

Theorem 3: (Ore, 1960)

if G is simple graph with $(n \geq 3)$ vertices and $\deg(u) + \deg(v) \geq n$ for each pair of non-adjacent vertices u and v , then G is hamiltonian.

proof: we assume that it is false, and we derive contradiction, so let G be a non-hamiltonian graph with n vertices, satisfies the given condition on the vertex degrees. By adding extra edges if necessary, we may assume that G is 'only just' not-hamiltonian in the sense that the addition of further edge gives a hamiltonian graphs (note that adding extra edge doesn't violate the constraint of the degrees), it follows that if G contain a path $v_1 \rightarrow v_2 \rightarrow \dots \rightarrow v_n$ passing through every vertex. But since G is non-hamiltonian, the vertices v_1, v_n are not adjacent, and so $\deg(v_1) + \deg(v_n) \geq n$; by pigeonhole principle, there must be some vertex v_i adjacent to v_1 with the property that v_{i-1} is adjacent to v_n , but this gives the required contradiction since $v_1 \rightarrow v_2 \rightarrow \dots \rightarrow v_{i-1} \rightarrow v_n \rightarrow v_{n-1} \rightarrow \dots \rightarrow v_{i+1} \rightarrow v_i \rightarrow v_1$ is then hamiltonian cycle.



Corollary 7.2: (Dirac, 1952)

if G is a simple graph with $n \geq 3$ vertices, and if $\deg(v) \geq \frac{n}{2}$, for each vertex v , then G is Hamiltonian

proof: The result follows from precedent theorem since $\forall v, u \in V(G)$
 $\deg(v) + \deg(u) \geq n \Rightarrow G$ is Hamiltonian.

Theorem: Hamiltonian Digraph (H.A. Ghouila-Houri).

Let D be a strongly connected digraph with n vertices. If $\text{outdeg}(v) \geq \frac{n}{2}$ and $\text{indeg}(v) \geq \frac{n}{2}$ for each vertex v , then D is Hamiltonian.